MIL-R- 46341 (MU)
30 November 1962

MILITARY SPECIFICATION

RANGE FINDER, FIRE CONTROL: M14

1. SCOPE

1.1 This specification covers one type of stereoscopic instrument known as Range Finder: M14.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	-Optical Components of Fire Control Instruments: General Specification Governing the Manufacture and Inspection of
MIL-F-13926	-Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45208	-Inspection Requirements, General Specifications for

STANDARDS

Military

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	-Identification Marking of U. S. Military Property
MIL-STD-202	-Test Methods for Electronic and Electrical Component Parts

DRAWINGS

Munitions Command

F8224500	-Range Finder, Fire Control: M14

FSC 1240

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(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Initial production sample.- Unless otherwise directed by the contracting officer, an initial production sample shall be required. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Fabrication.- The range finder shall be manufactured in accordance with Drawing F8224500, "Range Finder, Fire Control: M14" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired optical, mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities.

3.4 General specifications.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specifications MIL-O-13830 and MIL-F-13926.

3.4.1 MIL-O-13830

(a) Cleanliness
(b) Illuminated reticle

3.4.2 MIL-F-13926

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Electrical
(f) Workmanship

3.5 Pre-Vibration.- The range finder prior to vibration shall be set up as shown on the assembled view Drawing F8224500 (sheet seven) with the mounting surface of the bearings horizontal and the following conditions met in sequence:

(1) Centerline of the mounting bearings horizontal
(2) End housing assemblies in place
(3) Boresight knobs set at "3" (normal setting)
(4) Ammunition scale set at "0"
(5) Ballistic correction knob set at "0"

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3.5 Pre-Vibration (Continued)

(6) Scales and reticles illuminated
(7) Scales projected into the left eyepiece
(8) Filters out of the optical system
(9) The bottom of the central ranging mark set at 7.0 plus or minus 0.5 mil above the center of the cross on the sighting reticle
(10) Elevation level bubble centered
(11) Provide a target at 484 to 525 yards with the range scale set to agree with the range chosen

3.5.1 Reticle alinement.- The line of sight of the left telescope through the boresight mark (center of cross of the sighting reticle) shall be horizontal within plus or minus 1 mil and shall be perpendicular to the centerline of the mounting bearing keys within plus or minus 1 mil. The line of sight of the right telescope through the boresight mark shall be such that the aiming point coincides with the aiming point of the left telescope with a maximum angular separation of 0.2 mil with and without filters under all conditions (see 3.8.16).

3.5.2 Collimation.- With the eyepieces set at infinity focus, the rays from the same point on the target emerging from the two eyepieces shall not diverge more than 30 minutes of arc horizontally nor more than 12 minutes of arc vertically and shall not converge horizontally. These conditions shall be met with the interpupillary knob set at all positions between 58 and 72 mm, at all elevations with and without filters under each condition.

3.5.3 Parallax of sighting reticle.- Provide a target of 1500 plus or minus 200 yards and set the range scale to agree with the range chosen. Parallax in the right and left optical systems at the optical axis (boresight mark) shall not exceed 0.1 mil (width of reticle line).

3.5.4 Parallax of stereoscopic reticle.- The change in the vertical and horizontal position of the observer's eyes to the extremes of the exit pupil shall not cause more than 5 Units of Error (UOE) median ranging error from the reading in the center of the exit pupils in the apparent relative distance of the target marks when ranging on a target at 1500 plus or minus 200 yards distance, with and without the filters. In addition, all median readings shall be within a spread of 6 UOE. Increasing the interpupillary distance setting by 2 mm shall not cause more than 2 UOE median ranging error in the apparent relative distance of the target marks when ranging on the above target both with and without the filters.

3.6 Diopter setting.- With the proper halving adjustment, the observer shall make 5 Internal Correction Settings (ICS) readings on the target specified in 3.5.3, recording the median of these 5 readings. The eyepiece shall then be set at zero diopters using a dioptometer; readings of the eyepiece scales shall be read and recorded within 0.1 diopter.

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3.7 Vibration.- The range finder with the end housings attached, mounted in its normal operating position shall be vibrated in a vertical plane at a constant frequency of thirty cycles per second with an amplitude of 1/16-inch (1/8-inch total excursion) for a period of five minutes plus or minus fifteen seconds. Subsequent to vibration there shall be no evidence of damage or physical failure and the range finder shall show no more than 5 particles of dirt 0.2 mil maximum real diameter in the outer 20° of the apparent field and no more than 4 particles of dirt 0.05 mil maximum real diameter in the central 17.5° of the apparent field viewed with the eyes at the exit pupils of the instrument. Bright spots (pinholes in reticles) shall be considered as dirt. Dirt on surfaces out of the focal plane shall not exceed in size and number the specifications for size and number of digs permitted for that surface.

3.8 Post vibration.- The range finder restored to the conditions of 3.5, the requirements listed below are allowed but the maximum tolerances of 3.5.1, 3.5.2, 3.5.3, and 3.5.4 shall not be exceeded:

(a) The line of sight of the sighting reticle shall not have changed by more than 0.2 mil due to vibration.
(b) Eyepiece collimation shall not have changed by more than 6 minutes in any direction after vibration.
(c) With the eyepiece scales restored to the setting recorded under 3.6, a dioptometer shall indicate a value less than plus or minus 0.2 diopter from that obtained prior to vibration.
(d) The halving adjustment shall not have changed by more than 1/4 the length of the central ranging mark of the stereoscopic reticle.
(e) The observer shall repeat the series of 5 ICS readings on the target specified in paragraph 3.5.3 and again record the median of these five readings. This value shall not vary by more than 3 UOE from those obtained before vibration.

3.8.1 Exit pupils.- Mininum diameters of the exit pupils shall be 4.5 millimeters when viewed in the direction of the optical axis.

3.8.2 Interpupillary adjustment.- The interpupillary scale shall indicate the correct interpupillary distance within 0.5 mm at any setting on the interpupillary scale.

3.8.3 Eyepiece focus.- Eyepiece focus for the ballistic and ranging reticles shall be within 0.25 diopter when the eyepiece diopter scale reads zero. The eyepiece shall be adjustable over the range of plus and minus 3 diopters, within 0.25 diopters, as read on the eyepiece scales.

3.8.4 Magnification.- The magnification of the left and right systems shall be equal within one percent.

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3.8.5 Resolution.- The resolving power on the optical axis of each optical system for a parallel bar test target shall be 6.0 seconds of arc or less.

3.8.6 Light transmission.- The difference in the light transmission between the two optical systems shall not exceed 20 percent of the higher transmission value.

3.8.7 Accuracy of sighting reticle.- The angular value of the total horizontal lines and spaces of the reticle, in either field of view, shall be 40, plus or minus 1 mil.

3.8.8 Parallax of ammunition and range scales.- Parallax between the index and range scale shall be not more than 1/2 the width of the index line; parallax between the index and ammunition scale shall be not more than twice the width of the index line.

3.8.9 Image tilt.- The images of a distant horizontal target line formed by the two optical systems shall be parallel within 30 minutes of arc and both images shall be horizontal within 1° with the level bubble centered.

3.8.10 Parallelism of sighting reticle and image.- The horizontal line of the sighting reticle, when projected into either eyepiece, shall be parallel to the image of a horizontal target line within plus or minus 43 minutes of arc.

3.8.11 Plumb travel.- The boresight mark, when projected into left eyepiece, shall not deviate more than 0.2 mil from a plumb line when the range finder is elevated and depressed through its full excursion.

3.8.12 Internal correction setting.- When the ICS knob, calibrated in UOE's, is moved from its midpoint to any other graduation the change in range (yards) measured in UOE's shall be equivalent to the corresponding change in number of UOE's on the knob within 15%. The scale of the ICS knob shall be adjusted to read 25 plus or minus 5 for the average observer. When the ICS scale is set to an increased reading, the range finder shall record longer ranges.

3.8.13 Parallelism of stereoscopic reticles and image.- The two stereoscopic ranging reticles shall be parallel to the image of a plumb line within 2.5 degrees of arc (1/4 line width over the length of the central range mark). Parallelism of the stereoscopic reticle marks shall be such that median range readings made at the top and at the bottom of the central range mark shall not differ by more than 2 Units of Error (UOE). Ranging first with a target adjacent to the right side of the central range mark, then adjacent to the left shall not cause a range error greater than 2 UOE.

3.8.14 Range error due to height of stereo reticle adjustment.- After moving the left and right reticles together vertically plus 3 and then minus 3 mils, the median of five ICS settings at both extremes must not differ from that of the nominal position by more than 5 UOE.

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3.8.15 Halving error.- When the range scale is actuated from 500 yards to 5000 yards, the dipvergence of the target in the right eyepiece shall not change by more than 2 minutes of apparent field between 500 to 900 yards, and between 900 and 5000 yards.

3.8.16 Filters.- Introduction of the filters into the optical system shall not cause more than 1 UOE in ICS, 0.1 mil halving error, 0.1 mil vertical boresight error, and 0.1 mil horizontal boresight error, all measured in the real field.

3.8.17 Range accuracy.- The range indicated on the range scale shall not differ from the true range by more than 2 UOE. One UOE in yards is 3.8R2, where "R" is the range in thousands of yards. (This formula is valid only for values up to and including 5 UOE's.)

3.8.18 Superelevation accuracy.- When ranges from 500 to 5000 yards are introduced in each direction through the range knob and indicated on the range scale in accordance with the tabulated values of Table I for various ammunition settings, the cam shall depress the line of sight through the required superelevation angle within the specified tolerances when top surfaces of the left mounting bearing housing as shown on Drawing F8242670-1 remains immobilized and is horizontal in two planes.

TABLE I

RANGE (SCALE SETTING) (Yards)	BALLISTIC Correction Knob	AMMUNITION (SCALE OR KNOB SETTINGS) 88	5	TOLERANCES MILS Plus or Minus
B	0	*	*	
500	0	3.8	0	.30
1000	0	8.0	3.4	.30
1500	0	12.3	5.4	.30
2000	0	17.0	7.3	.30
2000	+15	20.0	8.5	.30
2000	-10	15.2	6.6	.30
2500	0	22.1	9.2	.40
3000	0	27.5	11.2	.50
3000	+15	32.5	13.0	.50
3000	-10	24.6	10.1	.50
3500	0	33.0	13.2	.60
4000	0	37.7	15.3	.80
4500	0	46.2	17.5	1.00
5000	0	57.5	19.8	1.40

* Ammunition Scale at 0
Superelevation - 0.0

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3.8.19 Cant mechanism.

3.8.19.1 Cant control knob.- With the following conditions met, the Cant Control Knob A8242666 shall be capable of being rotated to the right and left of the index mark a maximum 2.75 plus or minus .01 revolutions:

(a) Cant correction level vial centered
(b) Ammo knob set at 88
(c) All conditions with exception of (4), (9) and (11) of paragraph 3.5
(d) Requirements in paragraph 3.5.1
(e) Location of Spiral Gear Segment B8224491 set up in accordance with note outlined on Drawing F8224490
(f) Cant control knob alined in accordance with note outlined on Drawing C8224506

3.8.19.2 Reticle movement.- With the range finder set up in accordance with paragraph 3.8.19.1, the reticle movement resulting from various angles of cant introduced by the rotation of cant control knob in each direction (clockwise and counterclockwise) at ranges set in as mils of superelevation shall be in accordance with values listed in Table II.

TABLE II

Reticle Movement in Mils

Cant Control Knob	Mils of Superelevation (Range)				
	8.0 (1000 yds)	17.0 (2000 yds)	27.5 (3000 yds)	37.7 (4000 yds)	53.3 (4818 yds)
2 Revolutions (11.25°)	1.6 ± .3	3.4 ± .4	5.4 ± .5	7.3 ± .6	10.3 ± .7
1 Revolution (5.625°)	.8 ± .3	1.7 ± .4	2.7 ± .5	3.7 ± .6	5.2 ± .7
0 Revolution	0 ± .25	0 ± .25	0 ± .25	0 ± .25	0 ± .25

NOTE: When the right hand side of the cant level vial bubble is elevated, the reticle shall move to the right and vice versa.

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3.8.19.3 Reticle position.- With conditions as specified in Table III and IV below, introduced separately into the range finder, and each followed by returning the range scale to (B), the ammunition and ballistic correction knobs to zero, and the cant correction to detent, the sighting reticle shall return to a common zero setting within 0.10 mils.

TABLE III	TABLE IV
Range 4000 yards (minimum) Ballistic Correction-Plus 10 (min.) Ammo - 80 Cant - 2 revolutions (min.) Counterclockwise	Range 4000 yards (minimum) Ballistic Correction-Minus 10(min) Ammo - 80 Cant - 2 revolutions (min.) Clockwise

3.8.20 The base length of the range finder measured optically, when the nominal certified end housings are installed shall be 73 plus or minus .040 inch.

3.8.21 Backlash.- Backlash in the boresight knobs shall not exceed 0.05 mil in the line of sight.

3.8.22 Knob travel.- All knobs shall have sufficient movement to cover the entire range of their corresponding scales.

3.8.23 Boresight knobs.- The running torque required to rotate the boresight knobs in each direction shall not exceed 10 pound inches.

3.8.23.1 Boresight knobs.- The locking of the boresight knobs shall not cause more than 0.05 mil shift in the line of sight and shall not cause more than 0.05 mil rotation of the knobs as measured on the knob scale.

3.8.23.2 Elevation boresight knob travel.- Elevation boresight knob shall have at least 5 mils deflection in depression and at least 4 mils in elevation.

3.8.23.3 Azimuth boresight knob travel.- Azimuth boresight knob shall have at least 4.5 mils deflection each side of normal position established in paragraph 3.5.1.

3.8.24 Range knob.- The running torque required to rotate the range knob in each direction shall not exceed 16 pound inches when the ammo and ballistic correction knobs are set to a maximum clockwise direction.

3.8.25 Correction knobs.- The running torque required to rotate the halving, ICS, ballistic correction and ammo knobs in each direction shall not exceed 8 pound inches. Ballistic correction knob shall be checked when range knob is rotated a maximum counterclockwise and the ammo knob a maximum clockwise. Ammo knob shall also be checked when range knob is rotated a maximum counterclockwise and ballistic knob a maximum clockwise.

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3.8.26 Interpupillary knob.- The running torque required to rotate the interpupillary knob in each direction shall not exceed 8 pound inches.

3.8.27 Diopter adjustments.- The diopter knobs shall be oscillated 10 times maximum displacement at room temperature. The running torque required to rotate the diopter adjustments in each direction after this operation shall not exceed 11 pound inches.

3.8.28 Elevation accuracy.- When Input Flange A7648968 is rotated in a clockwise direction without overtravel, the elevation of the range finder and Output Coupling B7649018 shall conform to the following table:

Revolutions of Input Flange A7648968	Elevation in Mils (Range Finder and Coupling B7649018)	Tolerance in Mils (Range Finder)	(Coupling B7649018)
0	0	.00	.00
1	64.0	±0.20	±0.30
2	128.0	±0.20	±0.30
3	192.0	±0.20	±0.30
4	256.0	±0.20	±0.30
5	320.0	±0.20	±0.30
6	384.0	±0.20	±0.30

3.8.28.1 Backlash.- When Flange A7648968 is rotated in a counter-clockwise direction without overtravel to the same settings as in 3.8.28 in order to determine the backlash of the system, the elevation values obtained shall not vary from the first set of values by more than 0.1 mil.

3.9 Sealing.- The range finder body and end housings shall show no evidence of leakage when subjected to an internal pressure of 6 inches of mercury at standard ambient temperature (plus 60 to 90 degrees Fahrenheit) for a period of twelve hours. Correction shall be made for any pressure variation resulting from temperature and barometric changes.

3.9.1 Flushing.- Upon satisfactory completion of the sealing requirements of 3.9, the range finder and the end housings shall be flushed with dry nitrogen. The flushing shall continue until the gaseous mixture bled from the range finder and the end housings have a dew-point of at least minus 25 degrees Fahrenheit. The instruments shall then be sealed with an internal pressure of dry nitrogen of between 0.2 and 0.5 inches of mercury.

3.10 Environmental requirements.

3.10.1 Storage temperatures.- The range finder with end housings shall be exposed to ambient temperatures of minus 80 and plus 160 degrees Fahrenheit until thermal equilibrium is reached. After thermal equilibrium has been re-established at standard ambient temperature, the range finder

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and end housings shall show no evidence of physical failure such as cement separation, bonding failure, glass breakage, casting failure, etc., and shall meet the requirements of 3.8 to 3.8.28.1 inclusive. Readjustment of the range finder will be permitted following storage temperatures, but will require that the range finder be completely rechecked for conformance with all post vibration requirements of 3.8 to 3.8.29 inclusive.

3.10.2 Operational temperatures.- The range finder with end housings shall be exposed to an ambient temperature of minus 40 degrees Fahrenheit until thermal equilibrium is reached, and inspected at that temperature to insure that all knobs, locks and operational adjustments are mechanically operable. The range finder with end housings shall then be exposed to a temperature of minus 25 degrees Fahrenheit until thermal equilibrium is reached and then shall be inspected at this temperature for conformance with the requirements of 3.8.23 to 3.8.27 inclusive. The range finder with end housings shall be exposed to an ambient temperature of plus 150 degrees Fahrenheit until thermal equilibrium is reached and inspected at this temperature for conformance to the requirements of 3.8.23 to 3.8.27 inclusive.

3.11 Sealing recheck.- Range finder with end housings which have met the environmental requirements of 3.10 shall be rechecked to the requirements of 3.9 for leakage and then be reflushed to meet the requirements of 3.9.1. Failure of any range finder or end housing to meet the requirements of 3.9 after completion of the environmental requirements of 3.10 shall not be cause for rejection of the unit provided the range finder or end housing is resealed to meet the requirements of 3.9 and 3.9.1.

3.12 Electrical check.- Electrical checks may be conducted at any time prior to final acceptance of the range finder.

3.12.1 Insulation resistance.- Insulation resistance between ungrounded circuits and the body or frame and between isolated circuits shall not be less than 2 megohms with 500 volts dc applied.

3.12.2 Continuity.- The electrical circuits shall be checked for continuity and shall conform to the wiring diagram.

3.13 Identification marking.- Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 General requirements.- Inspection requirements and procedures shall be in accordance with Specification MIL-I-45208.

4.3 Initial production sample.- Prior to production, the contractor shall submit the initial production sample in accordance with 3.1 for inspection by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 Production sample failure.- Should the production sample fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. Where corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.4 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.4.1 Defective units.- Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.5 Material.- Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.6 Inspection of components.- All positional and functional characteristics shown on drawings, or contained in notes on the drawings or referenced specifications, which indicate precision requirements and have a bearing on the functional accuracy and performance of the range finder shall be inspected in accordance with the written inspection system required by Specification MIL-I-45208. These characteristics are defined as requirements effecting concentricity, squareness, parallelism, micro-finishes 64 RMS or better, threads, identification marking and performance.

4.6.1 Sampling inspection.- Where quality control has been established on in-process manufacture, sampling inspection of components may be used in lieu of 100 percent inspection and shall be in accordance with Table V.

TABLE V - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

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4.6.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.7 Inspection equipment.- Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

4.7.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.8 Tests.

4.8.1 Classification of defects.- The classification of defects in Tables VI and VII shall constitute the minimum inspection requirements to be performed by the contractor.

4.8.1.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE VI - CLASSIFICATION OF DEFECTS

Use Inspection Level L-7 of Table III-B and Table IV-A of Standard MIL-STD-105

CRITICAL: None defined

MAJOR: AQL 6.5% Defective	Req't Par.	QA Par.	Test Method
101. Vibration (Cleanliness)	3.7	4.9.2	Special Test Equipment (STE)
102. Post vibration	3.8	4.9.3	STE
103. Boresight knobs	3.8.23	4.9.26	STE
104. Range knob	3.8.24	4.9.27	STE
105. Correction knobs	3.8.25	4.9.28	STE
106. Interpupillary knobs	3.8.26	4.9.29	STE
107. Diopter adjustments	3.8.27	4.9.30	STE
108. Sealing	3.9	4.9.33	STE

MINOR: None defined

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TABLE VII - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: None defined

MAJOR:	AQL 1.5 % Defective	Req't Par.	QA Par.	Test Method
101.	Reticle alinement	3.5.1	4.9.1.1	Special Test Equipment(STE)
102.	Collimation	3.5.2	4.9.1.2	STE
103.	Parallax, sighting reticle	3.5.3	4.9.1.3	STE
104.	Parallax, stereo reticle	3.5.4	4.9.1.4	STE
105.	Exit pupils	3.8.1	4.9.4	Standard Measuring Equipment (SME)
106.	Interpupillary adjustment	3.8.2	4.8.5	SME
107.	Eyepiece focus	3.8.3	4.9.6	SME
108.	Magnification	3.8.4	4.9.7	SME
109.	Resolution	3.8.5	4.9.8	SME
110.	Light transmission	3.8.6	4.9.9	STE
111.	Accuracy of sighting reticle	3.8.7	4.9.10	STE
112.	Parallax of ammunition and range scales	3.8.8	4.9.11	STE
113.	Image tilt	3.8.9	4.9.12	STE
114.	Parallelism of sighting reticle and image	3.8.10	4.9.13	STE
115.	Plumb travel	3.8.11	4.9.14	STE
116.	Internal correction setting	3.8.12	4.9.15	STE
117.	Range error	3.8.14	4.9.17	STE
118.	Halving error	3.8.15	4.9.18	STE
119.	Filters	3.8.16	4.9.19	STE
120.	Range accuracy	3.8.17	4.9.20	STE
121.	Superelevation accuracy	3.8.18	4.9.21	STE
122.	Cant control knob	3.8.19.1	4.9.22	STE
123.	Reticle movement	3.8.19.2	4.9.22	STE
124.	Reticle position	3.8.19.3	4.9.22.1	STE
125.	Base length	3.8.20	4.9.23	STE
126.	Backlash	3.8.21	4.9.24	STE
127.	Knob travel	3.8.22	4.9.25	Visual
128.	Boresight knob locks	3.8.23.1	4.9.26.1	STE
129.	Elevation boresight knob	3.8.23.2	4.9.26.2	STE
130.	Azimuth boresight knob	3.8.23.3	4.9.26.3	STE
131.	Elevation accuracy	3.8.28	4.9.31	STE
132.	Backlash	3.8.28.1	4.9.32	STE
133.	Electrical output	3.8.29	4.9.33	STE
134.	Insulation resistance	3.12.1	4.11.1	SME
135.	Continuity	3.12.2	4.11.2	SME
136.	Identification marking	3.13	4.12	Visual

MINOR: None defined

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4.9 Test method and procedures.

4.9.1 Pre-vibration.- The range finder shall be mounted on special test equipment which will allow it to meet the conditions specified in 3.5. A line of sight through the range finder shall be directed to a target, positioned at the specified distance, appropriate for measuring the accuracy requirements of the range finder. This target shall be positioned perpendicular with respect to the range finder mounting surfaces. All settings shall be made as specified in 3.5.

4.9.1.1 Reticle alinement.- This test shall be conducted with the range finder set up as in 4.9.1 and the sighting reticle projected into the left optical system. The movement necessary to coincide the boresight mark of the righting reticle with a boresight point on the target, as measured on the elevation and azimuth boresight knobs, shall not exceed the tolerance specified in 3.5.1. With the sighting reticle projected into the right optical system, the deviation of the right line of sight from the left line of sight, through the boresight mark, shall meet the requirements of 3.5.1.

4.9.1.2 Collimation.- The range finder shall be set up as in 4.8.1. This test shall be conducted with the aid of an auxiliary testing device, positioned at the range finder eyepieces, capable of combining the target images emerging from the eyepieces onto a reticle appropriate for measuring divergence (displacement in azimuth between two images) and dipvergence (displacement in elevation between two images). Divergence and dipvergence shall be read with the IPD adjustment set at the median and two extreme positions of the IPD scale when the line of sight through the range finder is horizontal, minus 5 degrees and plus 15 degrees. The divergence and dipvergence of the range finder shall meet the requirements of 3.5.2.

4.9.1.3 Parallax of sighting reticle.- This test shall be conducted with the sighting reticle projected into each optical system in turn. Parallax at the boresight mark for each system shall not exceed the tolerance specified in 3.5.3.

4.9.1.4 Parallax of stereoscopic reticle.- This test shall be conducted with the aid of an auxiliary testing device, positioned at the range finder eyepieces, containing two (2) apertures accurately calibrated for the observer's interpupillary distance. All rangings to a target positioned at 1500 yards shall be accomplished by means of the internal correction setting (ICS) knob. The median range readings made at the center, the two (2) horizontal and two (2) vertical positions of the exit pupil shall meet the requirements of 3.5.4. Increasing the PID setting of the range finder eyepiece specified distance shall not exceed the tolerance specified in 3.5.4.

4.9.1.4.1 Filters.- The introduction of filters into the optical systems shall not cause the median range reading at the center of the exit pupil to exceed the tolerance specified in 3.5.4.

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4.9.1.4.2 Interpupillary adjustment.- The interpupillary adjustment shall be inspected at the median ("65" IPD) and two extremes of the IPD scale. This test shall be conducted with the aid of an auxiliary testing device, positioned at the range finder eyepieces, capable of combining the target images emerging from the eyepieces into a single eyepiece. This device shall also have a means of accurately measuring interpupillary distance of these two emerging target images. With the etched dots on objective lenses Drawing A7632698 in focus at the testing device eyepiece, the etched dots shall be brought into coincidence by means of the interpupillary adjustment of the testing device. The interpupillary distance set on the range finder under IPD scale shall not differ from the IPD value established on the testing device by more than the tolerance specified in 3.6.

4.9.2 Vibration.- The range finder shall be mounted in its normal operating position on a testing device and vibrated in accordance with the requirements of 3.7. Subsequent to vibration it shall be inspected for cleanliness in accordance with Specification MIL-O-13830 and shall meet the applicable requirements of 3.7.

4.9.3 Post vibration.- The range finders that have been vibrated shall be remounted in special test equipment and all settings again be made as specified in 3.5. The following requirements under 3.8 shall be checked and must not exceed the tolerance specified.

4.9.4 Exit pupil.- This test shall be conducted in accordance with Specification MIL-O-13830 by using standard measuring equipment and shall meet the requirements of 3.8.1.

4.9.5 Interpupillary adjustment.- The interpupillary distance shall be inspected using standard measuring equipment and shall be within the tolerances specified in 3.8.2.

4.9.6 Eyepiece focus.- The diopter movement of each eyepiece shall be tested manually and visually for the requirements of 3.8.3. Focus of each eyepiece shall be tested with the aid of a calibrated dioptometer having a magnification of at least three power (3X) and adjusted for infinity focus for the observer's eye. The zero setting of each eyepiece diopter scale shall be within the tolerance specified in 3.8.3 when sharp focus of the sighting reticle is obtained.

4.9.7 Magnification.- This test shall be conducted in accordance with Specification MIL-O-13830 by using standard measuring equipment and shall meet the requirements of 3.8.4.

4.9.8 Resolution.- This test shall be conducted in accordance with Specification MIL-O-13830. The resolving power of each system shall meet the requirements of 3.8.5.

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4.9.9 Light transmission.- Filters shall be transferred out of the optical systems for this test. Light rays from a concentrated source entering the range finder through each end housing window shall be measured at each eyepiece with special test equipment appropriate for measuring light transmission. Light transmission through the two optical systems shall meet the requirements of 3.8.6.

4.9.10 Accuracy of sighting reticle.- This test shall be conducted in accordance with Specification MIL-O-13830. The angular substance of the total horizontal lines and spaces of the reticle shall meet the requirements of 3.8.7 when the sighting reticle is projected into each optical system.

4.9.11 Parallax of ammunition and range scales.- Parallax between the index and range scale shall be checked by means of special testing equipment and shall meet the requirements of 3.8.8.

4.9.12 Image tilt.- This test shall be conducted with the aid of an auxiliary testing device capable of combining the target images on to a reticle suitable for measuring image tilt of the two optical systems. Image tilt of the two optical systems shall meet the requirements of 3.8.9 when the elevation level bubble is centered in both a clockwise and counterclockwise direction.

4.9.13 Parallelism of sighting reticle and image.- Parallelism of the horizontal reticle line to the image of a horizontal target line shall be measured on a target appropriate for measuring any cant of the reticle line. Parallelism of the sighting reticle shall meet the requirements of 3.8.10.

4.9.14 Plumb travel.- A line of sight through the boresight mark of the sighting reticle coincident to a plumb line shall not deviate from the plumb line more than the tolerance specified in 3.8.11 when the range finder is elevated from maximum depression to maximum elevation by means of the manual elevation knob.

4.9.15 Internal correction setting.- The line of sight of the range finder shall be directed to a target of known distance within 1300 to 1700 yards. Adjustment of the ICS knob scale shall be tested by means of five (5) ICS readings to the target. The median of these five (5) readings shall fall within the tolerance specified in 3.8.12. When the ICS knob is rotated to one (1) extremity of the scale from the median ICS setting, the corresponding change in range, as read on the range scale when the stereo reticles have been returned to stereoscopic contact with the target by means of the ranging knob, shall be equivalent to the change in the number of units of error (UOE) within the specified tolerance. This test shall be repeated with the ICS knob rotated to the other extremity of the scale.

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4.9.16 Parallelism of stereoscopic reticle.- When one edge or corner of the central ranging mark of the left stereo reticle is alined to a vertical target line, the central ranging mark shall be parallel to the target line within the tolerance specified in 3.8.13. Two and one-half degrees (2.5^{o}) is equivalent to one-fourth (1/4) the width of the central ranging mark over its length. This test shall be repeated on the right stereo reticle. The difference between the two median ICS readings taken with the central ranging mark positioned immediately above, then immediately below a point on the target shall not exceed the tolerance specified in 3.8.13.

4.9.17 Range error due to stereo reticle movement.- The line of sight of the range finder shall be directed to a target appropriate for ranging with the stereo reticles. Median ICS readings shall be taken with the stereo reticles in their nominal position, then 3 mils above, then 3 mils below the nominal position. The difference between the median at each extreme and the median at the nominal shall not exceed the tolerance specified in 3.8.14. Movement of the right stereo reticle shall be accomplished by rotation of the recessed adjustment screw located in the bottom, left side, of the main housing. Movement of the left reticle shall be accomplished by rotation of the halving knob. Caution shall be exercised to return the stereo reticles to original position after test.

4.9.18 Halving error.- The sighting reticle shall be projected into the right optical system and a line of sight through the reticle boresight mark shall be directed to a horizontal target line. Deviation of the line of sight with respect to the horizontal target line shall not exceed the tolerance specified in 3.8.15 when the range scale is actuated from 500 to 5000 yards.

4.9.19 Filters.- Introduction of the filters into the optical systems shall not cause more errors than those specified in 3.8.16 when measured in the real field.

4.9.20 Range accuracy.- The ICS knob shall be set to the observer's median ICS reading and the line of sight directed to a target or targets positioned at known distances between 500-600 yards, 600-1100 yards, 1100-1700 yards, 1700-2200 yards, and 2200-5000 yards. Five range readings shall be made to each target. The median range readings of each group of five rangings shall meet the requirements of 3.8.17 by application of the given formula; i.e. to find one UOE for 1500 yards of range, apply formula:

$$3.4\ R^2 = 3.4\ \frac{1500^2}{1000} = 3.4\ (1.5)^2 = 3.4 \times 2.25 = 7.65 \text{ yards}$$

2 UOE for this range = 7.65 x 2 = 15.3 yards

4.9.21 Superelevation accuracy.- With the range finder mounted in special test equipment, the required ranges shall be set at the 88 ammo setting and then the 5 ammo setting. At each of these settings the reading taken on a target, which is part of the special test equipment, shall not exceed the tolerance specified in the table of 3.8.18.

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4.9.22 Cant mechanism.- Rotation of the cant control knob shall be visually and manually inspected for the requirements of 3.8.19. Rotation of the cant control knob in a clockwise direction shall elevate the right side of the cant level vial. Azimuth movement of the sighting reticle resulting from angles of cant introduced by rotation of the cant control knob in each direction, clockwise and counterclockwise from detent position shall meet the requirements of Table II of 3.8.19.2. The test for cant correction shall be performed with a line of sight, through the boresight mark of the sighting reticle, directed to a target appropriate for measuring azimuth movement of the reticle.

4.9.22.1 Reticle position.- The tests specified in 3.8.19.3 shall be conducted under the conditions specified. The sighting reticle shall return to a common zero setting within the tolerance specified.

4.9.23 Base length.- The base lengths shall be measured by means of special testing equipment and shall meet the tolerance specified in 3.8.20.

4.9.24 Backlash.- Rotate a line of sight through the boresight mark of the sighting reticle, in a continuous clockwise direction by means of the azimuth boresight knob until the vertical reticle line is in coincidence with a vertical target line. Record reading on micrometer scale. Rotate line of sight past the vertical target line (clockwise) a minimum of one full turn of the azimuth boresight knob, then by a continuous counterclockwise rotation of the azimuth knob bring the vertical reticle line into coincidence with the vertical target line and read the micrometer scale. The difference between the recorded reading and second reading is the amount of backlash. Repeat this operation for the elevation boresight knob using the horizontal reticle line and a horizontal target line. Backlash for each knob shall not exceed the limit specified in 3.8.2.1.

4.9.25 Knob travel.- All knobs shall be visually and manually inspected for conformance with 3.8.22.

4.9.26 Boresight knobs.- The functional test for the boresight knob shall be performed with special testing equipment affixed to the boresight knobs that shall be capable of measuring running torque. The measured torque shall not exceed the amount specified in 3.8.23.

4.9.26.1 Boresight knob locks.- With a line of sight through the boresight mark of the sighting reticle, actuation of the boresight locks shall meet the requirements of 3.2.3.1.

4.9.26.2 Elevation boresight knob travel.- The line of sight shall be oriented as specified in 4.9.1. Rotation of the elevation boresight knob shall meet the requirements of 3.8.23.2.

4.9.26.3 Azimuth boresight knob travel.- The line of sight shall be oriented as specified in 4.9.1. Rotation of the azimuth boresight knob shall meet the requirements of 3.8.23.3.

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4.9.27 Range knob.- The specified torque required by 3.8.24 shall be measured by means of special test equipment and shall not exceed the tolerance specified.

4.9.28 Correction knobs.- The specified torque required by 3.8.25 shall be measured by means of special test equipment and shall not exceed the tolerance specified.

4.9.29 Interpupillary knob.- The specified torque required by 3.8.26 shall be measured by means of special test equipment and shall not exceed the tolerance specified.

4.9.30 Diopter adjustments.- The specified torque required by 3.8.27 shall be measured by means of special test equipment and shall not exceed the tolerance specified.

4.9.31 Elevation accuracy.- A line of sight through the range finder shall be directed to a target appropriate for measuring elevation accurately of the range finder. The sighting reticle boresight mark shall be alined to a boresight point on the target when the cant and elevation level bubbles are centered. The zero elevation position in the table shown in 3.8.2.8 will be established when the line of sight is depressed 128 mils below the horizontal with backlash eliminated from the input flange for rotation in a clockwise direction. Rotation of the input flange in a clockwise direction shall meet the requirements of the table shown in 3.8.28.

4.9.32 Backlash.- Rotation of the input flange in a counterclockwise direction, with backlash included, shall again meet the requirements of the table shown in 3.8.28 within the tolerance specified in 3.8.28.1.

4.9.33 Sealing of the range finder shall be tested with special test equipment that shall be capable of measuring any leakage of dry nitrogen from the range finder when pressurized as specified. This test shall meet the requirements of 3.9.

4.10 Environmental tests.

4.10.1 Storage temperatures.- After thermal equilibrium has been reached at the extreme temperatures specified in 3.10.1, the range finder shall be returned to standard ambient temperature and examined for any physical failure of optical and mechanical parts. The range finder shall then be inspected in accordance with tests specified in 4.9.1 to 4.9.33 inclusive.

4.10.2 Operational temperatures.- The range finder shall be exposed to an ambient temperature of minus 40 degrees Fahrenheit until thermal equilibrium is reached, and while at this condition the knobs, locks and operational adjustments shall be manually operated to ascertain if the requirements of 3.10.2 are met. The range finder shall then be exposed to an ambient temperature of minus 25 degrees Fahrenheit until thermal equilibrium

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is reached and then shall be inspected at this temperature in accordance with 4.9.26 to 4.9.30 inclusive. The range finder shall then be exposed to an ambient temperature of plus 150 degrees Fahrenheit until thermal equilibrium is reached and inspected in conformance with 4.9.26 to 4.9.30 inclusive.

4.11 Sealing recheck.- Subsequent to the environmental test the range finders which were subjected and have passed the environmental tests in 4.9 shall be rechecked for leakage and then reflushed in accordance with 3.9.1.

4.12 Electrical checks.

4.12.1 Insulation resistance.- This test shall be conducted in accordance with Standard MIL-STD-202 and shall meet the requirements of 3.12.1.

4.12.2 Continuity.- This test shall be performed with the range finder in operational condition. The electrical circuits shall be checked for continuity and shall conform to the wiring diagram.

4.13 Identification marking.- Marking shall be visually inspected for conformance of Standard MIL-STD-130.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with instructions forming part of the contract or furnished by the Contracting Officer in connection with the contract.

6. NOTES

6.1 Intended use.- The range finder is essentially an internally collimated, seventy-three (73) inch stereoscopic instrument incorporating two sighting telescopes and ballistics computing and transmitting mechanisms for use in M103, 120 MM Gun Tank.

6.2 Ordering data.- Invitation for bids and contracts shall specify the title, number and date of this specification and type of packing.

6.3 Definitions.

6.3.1 Median.- The term "median" as used herein, is defined as that range or setting which divides the series into an upper and lower half. For example, in a set of five range readings, 2000, 2050, 2080, 2060, 2025 the median would be 2050. Two ranges are higher and two are lower.

6.3.2 Standard test equipment (STE).- Special testing equipment is defined as special inspection fixtures, instruments, and modified standard measuring equipment. Appearance of STE in the specification does not imply that the equipment is available, or that a suitable design for such equipment is currently in existence.

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6.3.3 Standard measuring equipment (SME).- Standard measuring equipment includes the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

6.4 Elevating shaft assembly.- The elevating Shaft Assembly 7649023 for the range finder shall be mounted with Coupling 7654905 engaged in an immovable drive flange contained in a fixture representative of the drive flange of the Tank M103. The hole and slot of the flange shall have been machined to maximum allowable drawing tolerance. The drive pins of the clutch assembly of the opposite end of the elevating shaft shall be engaged in a fixture representative of the flange of the range finder. The hole and slot of the flange shall have been machined to maximum allowable drawing tolerance. The flange representative of the range finder shall be designed for rotation by application of a torque wrench. With the elevating shaft assembly mounted as described above, push the clutch assembly to compress the spring to bring the guide pin midway in the elongated slot. With a torque of 15 pound inches applied to the flange representative of the M14 range finder, in a clockwise and then a counter-clockwise direction, the total movement of the flange (the sum of the clockwise and counterclockwise movement) shall not exceed 0.030 of an inch as read at a 6-inch radius. (0.030 of an inch movement on a 6-inch radius is equal to 0.05 mil in the line of sight of the range finder).

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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